TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	85.0 ± 1.95
Correlation Stability	89.8 ± 0.40
Distribution Consistency	85.8 ± 3.57
Entropy Preservation	78.9 ± 0.70
Mean Z Score Drift	0.3 ± 0.08
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.9 ± 0.30
Feature Accuracy	85.0 ± 2.97
Novelty Score	0.8 ± 0.02

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Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	49.177	-11.657	19.2%	Warn
anaemia	0.431	0.464	+0.033	7.6%	OK
creatinine_phosphokinase	581.839	122.142	-459.697	79.0%	Drift
diabetes	0.418	0.510	+0.092	22.1%	Warn
ejection_fraction	38.084	40.808	+2.724	7.2%	OK
high_blood_pressure	0.351	0.448	+0.097	27.5%	Drift
platelets	263358.029	296875.423	+33517.393	12.7%	Warn
serum_creatinine	1.394	1.409	+0.015	1.1%	OK
serum_sodium	136.625	139.975	+3.349	2.5%	OK
sex	0.649	0.607	-0.042	6.5%	OK
smoking	0.321	0.347	+0.026	8.2%	OK
time	130.261	157.971	+27.710	21.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	85.0 ± 1.95
Correlation Stability (*2)	89.8 ± 0.40
Distribution Consistency (*3)	85.8 ± 3.57
Entropy Preservation (*4)	78.9 ± 0.70
Mean Z Score Drift (*5)	0.3 ± 0.08
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.9 ± 0.30
Feature Accuracy (*10)	85.0 ± 2.97
Novelty Score (*11)	0.8 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8066666666666666	0.635	0.7233333333333334	-0.08333333333333334	*
F1	0.6937863401621888	0.03574074074074074	0.42957693145908243	-0.26420940870310633	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7883333333333333	0.6050000000000001	0.7316666666666666	-0.05666666666666668	*
F1	0.675837074942099	0.24677082200397393	0.5655759112665135	-0.11026116367558556	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	0.61	0.7866666666666666	-0.04833333333333332	*
F1	0.7547418227770806	0.2702010462693144	0.6499926251965318	-0.10474919758054888	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8316666666666667	0.6383333333333333	0.7866666666666666	-0.045	*
F1	0.7556715392060958	0.2198013839323243	0.6469949477584623	-0.10867659144763346	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	0.6383333333333333	0.7716666666666667	-0.05333333333333336	*
F1	0.7417404771442427	0.1804995547419972	0.5955743336302626	-0.1461661435139802	*

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	0.6183333333333334	0.6166666666666668	0.00333333333333327	
F1	0.007142857142857143	0.12232289769054475	0.04514655172413793	0.03800369458128079	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	0.6433333333333333	0.6916666666666667	-0.05833333333333336	*
F1	0.5667171896628354	0.160681260101796	0.36159587485393935	-0.20512131480889595	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5316666666666666	0.5283333333333333	0.5549999999999999	0.02333333333333324	
F1	0.27138048447451885	0.19759663014144352	0.22457179463787876	-0.04680868983664024	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	0.6333333333333334	0.6333333333333334	0	
F1	N/A	0.0	N/A	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8366666666666667	0.6399999999999999	0.7883333333333333	-0.04833333333333336	*
F1	0.7454897762794181	0.12330006591686235	0.6289441013193827	-0.11654567496003558	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	35.8109698804418 7	0.0	152.0	0.0	43.0	1.0	213361.0	0.69004653231777 4	134.0	0.0
2	42.9237847290201 5	0.0	-59.0	1.0	38.0	1.0	336270.0	1.34045498177250 82	141.0	0.0
3	52.2949793864905 8	0.0	107.0	0.0	75.0	0.0	144600.0	0.82243527542528 18	142.0	1.0
4	49.7200218763309 04	1.0	407.0	1.0	36.0	0.0	353735.0	1.03660889351770 47	142.0	1.0
5	63.7173303263425 4	0.0	98.0	0.0	41.0	0.0	441108.0	0.73866698351592 83	143.0	0.0
6	42.8231150396260 1	1.0	307.0	0.0	71.0	1.0	334647.0	0.88900645110642 74	148.0	0.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	33.7469830440979 8	0.0	3899.0	1.0	39.0	0.0	292967.0	1.05652301430971 46	145.0	0.0
2	47.5442561884518 8	1.0	191.0	1.0	36.0	0.0	143038.0	1.10299027981313 77	130.0	0.0
3	42.6925568513140 45	0.0	770.0	1.0	45.0	0.0	154051.0	1.08701801724882 66	132.0	0.0
4	40.0112525661017 24	0.0	-137.0	1.0	64.0	0.0	117987.0	1.40554518859497 81	141.0	1.0
5	46.1899890651821 7	1.0	508.0	0.0	38.0	1.0	129209.0	1.42189663262326 45	142.0	1.0
6	32.8510364466497 7	1.0	459.0	0.0	42.0	1.0	29121.0	1.58231055176442	144.0	1.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	80.0577888810165 6	1.0	376.0	1.0	43.0	1.0	202932.0	1.50946928440625 18	131.0	0.0
2	97.9309772477225 4	0.0	264.0	0.0	43.0	1.0	241165.0	1.58974573069544 06	129.0	0.0
3	73.3350154819416 1	0.0	1160.0	0.0	32.0	0.0	283848.0	4.70939773878119 9	127.0	0.0
4	68.7408605886257 1	0.0	503.0	0.0	33.0	1.0	285163.0	0.71699805181216 49	134.0	0.0
5	73.5463053997740 8	1.0	429.0	0.0	42.0	1.0	254235.0	0.88765068577366 43	120.0	0.0
6	39.2756669153347 7	0.0	467.0	0.0	69.0	1.0	421225.0	0.75625274377823 46	127.0	1.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	66.4719226250122 7	0.0	98.0	0.0	58.0	0.0	331179.0	0.56661184195739 07	150.0	1.0
2	53.3520563973241 8	1.0	111.0	0.0	28.0	1.0	341272.0	1.17904767486746 7	131.0	1.0
3	61.5023081212547 1	1.0	1.0	0.0	64.0	0.0	457197.0	1.42073347472338 8	147.0	1.0
4	82.024158755413	0.0	890.0	1.0	43.0	1.0	203917.0	0.96315032259877 84	136.0	0.0
5	56.4329060781556 4	0.0	135.0	0.0	43.0	0.0	423862.0	1.93726511877065 02	140.0	0.0
6	53.6980542644981 1	1.0	-122.0	1.0	42.0	0.0	195110.0	1.22256915212524 58	136.0	1.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	60.3443136208626 1	1.0	5.0	0.0	34.0	0.0	364445.0	1.07786879804844 52	142.0	0.0
2	53.2883994114797 5	0.0	359.0	0.0	29.0	1.0	121046.0	0.63768396761538 16	138.0	1.0
3	61.6233779020216 6	0.0	-30.0	1.0	37.0	0.0	96963.0	0.19590801781481 315	141.0	1.0
4	60.7231521445117 15	0.0	617.0	0.0	44.0	0.0	193360.0	0.59742664765998 43	146.0	1.0
5	67.9424084058463 1	1.0	40.0	0.0	51.0	0.0	136472.0	1.56658473633692 63	143.0	1.0
6	53.5435182038526 9	1.0	1085.0	0.0	19.0	0.0	142242.0	1.23540439225188 47	144.0	0.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	41.4978695919057 6	0.0	238.0	0.0	44.0	0.0	154872.0	1.24619904754137 73	139.0	1.0
2	55.8748979282726 5	0.0	145.0	1.0	26.0	0.0	127496.0	2.80656484147496 5	137.0	1.0
3	68.6824835309603 4	0.0	864.0	0.0	43.0	1.0	301407.0	0.86845740192192 66	129.0	1.0
4	78.706594023614	1.0	177.0	1.0	45.0	0.0	146129.0	0.71871034785506 76	146.0	1.0
5	75.9526517065132	1.0	-64.0	0.0	32.0	0.0	222925.0	1.27669117340928 1	140.0	1.0
6	62.7299601106755 56	1.0	284.0	0.0	36.0	1.0	21367.0	0.95119602066878 36	130.0	0.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	48.676791092403036	0.0	1239.0	0.0	38.0	0.0	302606.0	1.4370880571616538	136.0	0.0
2	60.556558925847085	1.0	317.0	1.0	31.0	0.0	395955.0	0.821928136884022	137.0	0.0
3	57.866581889259315	1.0	105.0	0.0	45.0	0.0	361015.0	1.205117206585312	145.0	0.0
4	43.241617267719888	0.0	892.0	0.0	28.0	0.0	551439.0	0.7814737475472737	149.0	1.0
5	41.833313645926054	1.0	180.0	0.0	25.0	0.0	306121.0	1.1717388016894952	148.0	0.0
6	36.709778201714733	1.0	1306.0	0.0	13.0	0.0	265416.0	1.6854042688015504	144.0	1.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	95.430706834085537	0.0	1104.0	1.0	53.0	1.0	215910.0	0.8899753808665537	128.0	1.0
2	52.426136985133632	1.0	153.0	1.0	44.0	0.0	712807.0	1.5627882190515567	130.0	1.0
3	45.549920475130804	0.0	875.0	1.0	69.0	0.0	287351.0	1.2090266298492665	131.0	0.0
4	43.205993239202705	1.0	1452.0	0.0	35.0	0.0	228770.0	1.0489359781313565	140.0	0.0
5	55.930346897094377	1.0	1186.0	1.0	40.0	1.0	183331.0	2.654763792813864	132.0	1.0
6	69.364540312033604	1.0	992.0	0.0	49.0	0.0	266934.0	1.3010318721063785	134.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	35.37053332918675	0.0	108.0	0.0	22.0	1.0	15923.0	0.9815515069398479	143.0	1.0
2	67.90037593600941	1.0	240.0	0.0	17.0	0.0	208086.0	0.4961289073696499	134.0	1.0
3	66.66838535750334	0.0	1128.0	1.0	64.0	0.0	24592.0	0.6566475664068606	148.0	1.0
4	60.57279139219416	0.0	236.0	0.0	27.0	0.0	462951.0	0.4182622904440456	147.0	0.0
5	59.848929277229	1.0	3447.0	0.0	11.0	0.0	266209.0	0.8824190612790497	139.0	1.0
6	32.90087668884293535	0.0	30.0	0.0	27.0	0.0	100824.0	1.0526994461687895	138.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.48491685268874	0.0	259.0	0.0	43.0	1.0	54686.0	1.2535507511880932	132.0	1.0
2	58.94300653179506	1.0	3960.0	0.0	40.0	0.0	304926.0	1.467516892621383	136.0	0.0
3	52.64388098883926	0.0	104.0	0.0	39.0	0.0	435977.0	1.5874180671708187	139.0	0.0
4	62.0665263906617	1.0	564.0	1.0	32.0	0.0	-19916.0	1.1829546187584432	131.0	1.0
5	46.12597476387704	0.0	-3.0	0.0	52.0	0.0	103210.0	0.6819921682891652	136.0	1.0
6	74.2419755704103	0.0	237.0	0.0	43.0	0.0	19440.0	3.7194170036270893	143.0	1.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0